

Unraveling loose packets

Level 3: Step 1
HARD
Theme: Networked Systems: DNS & DHCP
Due Date: 12/12/2022

Please take the following notes in consideration:

- In this assignment the questions marked with a (*) are practical and have to be shown to the teacher (no report is needed).
- The remaining questions are to be answered in a written report, to be delivered to the teacher.
- Whenever possible, the VMs necessary to resolve the practical questions should be created as *linked clones*.

DNS and DHCP

1. When using the **BIND9 DNS Server**, Forward and Reverse Lookup zones are stored in local files called **zone files**. Investigate and describe their syntax giving an example of both for an hypothetical `linux.local` domain.
2. Investigate the **ISC DHCP Server** support for Standard DDNS (Dynamic DNS) Updates and provide a brief description of how it works.
3. In what circumstances and how is a zone transfer operated between **BIND9** primary/secondary servers?
4. (*) Considering the DNS domain `linux.local`, with the hosts:
 - `13s1h-dns1.linux.local` (Debian Linux 11) - DNS server (primary/master);
 - `13s1h-dns2.linux.local` (Debian Linux 11) - DNS server (secondary/slave);
 - `13s1h-dhcp1.linux.local` (Debian Linux 11) - DHCP server (main);
 - `13s1h-dhcp2.linux.local` (Debian Linux 11) - DHCP server (failover);
 - `13s1h-wstation.linux.local` (Ubuntu Desktop) - Workstation.

(a) Create a VM for each host in the **NAT network**. Make sure their hostnames are correctly set (see the above host list).

Start one of the hosts and take note of the **subnet, netmask, gateway and dns server** that were auto-configured.

Let's assume, for the remainder of the question, that these were retrieved:

 - **subnet/netmask** - 192.168.122.0/255.255.255.0
 - **gateway** - 192.168.122.2
 - **dns server** - 192.168.122.2

These are normally the ones in the default NAT network for **VMWare Workstation Pro**

(b) Modify the 4 Debian Linux hosts, changing their ip addresses to static:

- `13s1h-dns1.linux.local` - 192.168.122.200
- `13s1h-dns2.linux.local` - 192.168.122.201
- `13s1h-dhcp1.linux.local` - 192.168.122.100
- `13s1h-dhcp2.linux.local` - 192.168.122.101 Replace 192.168.122 above with the subnet you retrieved in a)

(c) **Disable the internal DHCP server on the NAT network.**

⚠ Don't forget to re-enable DHCP once you finish/submit this step or future VMs might no work correctly! ⚠

(d) Install the **ISC BIND9 Server** (bind9 package) on the `13s1h-dns1.linux.local` and `13s1h-dns2.linux.local` VMs.

Configure the **bind9** service as a **primary/master** name server on `13s1h-dns1` for the `linux.local` domain with the following Resource Records (RR):

- Forward Lookup zone:
 - `13s1h-dns1.linux.local` with Address 192.168.122.200
 - `13s1h-dns2.linux.local` with Address 192.168.122.201
 - `13s1h-dhcp1.linux.local` with Address 192.168.122.100
 - `13s1h-dhcp2.linux.local` with Address 192.168.122.101
- Reverse Lookup zone:
 - 192.168.122.200 points to `13s1h-dns1.linux.local`
 - 192.168.122.201 points to `13s1h-dns2.linux.local`
 - 192.168.122.100 points to `13s1h-dhcp1.linux.local`
 - 192.168.122.101 points to `13s1h-dhcp2.linux.local`

Replace 192.168.122 with the subnet retrieved in a)

Configure the **bind9** service as a **secondary/slave** name server on the `13s1h-dns2` server for the `13s1h-dns1` server.

Hint: Make sure the group `bind` is allowed to write files in `/etc/bind`.

(e) Install the **ISC DHCP Server** (isc-dhcp-server package) on both `13s1h-dhcp1` and `13s1h-dhcp2` VMs. It's normal for it to fail starting before it is configured.

Configure and start the **dhcpd** primary server on the `13s1h-dhcp1` host to offer the following IP configuration on your subnet (ex: 192.168.122.*):

- **domain name:** `linux.local`
- **dhcp range:** 192.168.122.10 → 192.168.122.50
Or the subnet retrieved in a)
- **router:** 192.168.122.2
Or the gateway retrieved in a)
- **domain name server:** 192.168.122.2
Or the dns server retrieved in a)

(f) Make the necessary changes to the `13s1h-dhcp1` server configuration so that `13s1h-wstation` is offered the 192.168.122.90 fixed IP address.

Also enable **DDNS Updates** so that the **DHCP Server** updates the record for `13s1h-wstation` in the `13s1h-dns1` server.

- (g) Configure and start the **dhcpcd** service on the `13s1h-dhcp2` server as **failover peer** for the `13s1h-dhcp1` server.
- (h) On the Ubuntu Desktop host (`13s1h-wstation`) set the **TCP/IPv4** configuration to obtain the IP and DNS addresses automatically. Confirm that the TCP/IP configuration was successful and the **subnet, netmask, gateway, dns server and domain name** are correct.
- (i) Prove that the DNS Service installed in d) is working, making use of the `nslookup` command (in the command prompt) to resolve the following names:
 - `13s1h-dhcp1`
 - `13s1h-dns1`
 - `13s1h-wstation`
 - `192.168.122.100`
Replace 192.168.122 above with the subnet you retrieved in a)
 - `www.google.com`
- (j) Stop the **bind9** service on the `13s1h-dns1` server and prove that the `13s1h-dns2` is responding to the same dns queries made in i).
- (k) Stop the **dhcpcd** service on the `13s1h-dhcp1` server and prove that the `13s1h-dhcp2` is responding to dhcp requests.

◇ Keywords ◇

- domain name service
- isc bind
- isc dhcp
- bind slave debian
- dhcp failover pool
- bind zone file syntax
- debian dhcp configuration

Deliverables

- 5. All deliverables submitted via [discord#assignments-submission](#)):
 - 1. PDF file with the report
 - 2. The file name should be `13s1h_aXXXXX_aYYYYY.pdf`. This code means:
 - Level 3
 - Step 1
 - Easy difficulty
 - Students that participated in the group
 - **Maximum:** 2 students